

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Supplementary Winter Examination – 2024

Course: B.Tech

Branch : Chemical Engineering

Semester : VI

Subject Code & Name: BTCHC603 & Process Instrumentation and control.

Max Marks: 60

Date:23/12/2024

Duration: 3 Hr.

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

		(Level/CO)	Marks				
Q. 1	Objective type questions. (Compulsory Question)		12				
1	The transfer function of a system is defined as:		1				
	<table border="1"> <tr> <td>a. Input/output ratio in time domain</td> <td>b. Input/output ratio in frequency domain</td> <td>c. Output/input ratio in frequency domain</td> <td>d. Output/input ratio in Laplace domain</td> </tr> </table>	a. Input/output ratio in time domain	b. Input/output ratio in frequency domain	c. Output/input ratio in frequency domain	d. Output/input ratio in Laplace domain		
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2	Stability of a control system can be analyzed using:		1				
	<table border="1"> <tr> <td>a. Bode plot</td> <td>b. Nyquist plot</td> <td>c. Root locus</td> <td>d. All of the above</td> </tr> </table>	a. Bode plot	b. Nyquist plot	c. Root locus	d. All of the above		
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3	Which of the following is the primary objective of process control?		1				
	<table border="1"> <tr> <td>a. Increase production cost</td> <td>b. Maintain process variables at desired values</td> <td>c. Decrease product quality</td> <td>d. None of the above</td> </tr> </table>	a. Increase production cost	b. Maintain process variables at desired values	c. Decrease product quality	d. None of the above		
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4	A PID controller includes which three control actions?		1				
	<table border="1"> <tr> <td>a. Predictive, Integral, Derivative</td> <td>b. Proportional, Integral, Diagnostic</td> <td>c. Proportional, Integral, Derivative</td> <td>d. Predictive, Interactive, Diagnostic</td> </tr> </table>	a. Predictive, Integral, Derivative	b. Proportional, Integral, Diagnostic	c. Proportional, Integral, Derivative	d. Predictive, Interactive, Diagnostic		
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5	What is the time constant (τ) in a first-order system?		1				
	<table border="1"> <tr> <td>a. Time to reach 63.2% of final</td> <td>b. Time to reach 100% of final</td> <td>c. Time to reach 50% of</td> <td>d. None of the above</td> </tr> </table>	a. Time to reach 63.2% of final	b. Time to reach 100% of final	c. Time to reach 50% of	d. None of the above		
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	value.	value	final value			
6	A Routh-Hurwitz criterion is used to:					1
	a. Analyze system dynamics	b. Tune PID controllers	c. Determine system stability	d. None of the above		
7	Which plot provides a graphical method to determine the phase margin?					1
	a. Root locus	b. Nyquist plot	c. Bode plot	d. None of the above		
8	The Ziegler-Nichols method is used for					1
	a. Measuring time constant	b. Controlling noise	c. Tuning PID controllers	d. Modeling nonlinear systems		
9	The Nyquist stability criterion is used to:					1
	a. Design control systems	b. Analyze frequency response	c. Determine system stability	d. None of the above		
10	The root locus method is used to:					1
	a. Analyze system stability	b. Measure steady-state error	c. Tune PID controllers	d. Both a and c		
11	What does a thermocouple measure?					1
	a. Pressure	b. Temperature	c. Flow rate	d. Level		
12	What type of controller is best for eliminating offset in steady-state errors?					1
	a. Proportional controller	b. Integral controller	c. On/off controller	d. Derivative controller		
Q. 2	Solve the following.					12
A)	What are thermocouples? Explain in details thermocouple wires and thermo wells.					6
B)	Explain Pressure measuring devices with neat and clear diagram.					6

Q.3	Solve the following.		12
A)	Derive the transfer function of second order system interacting liquid level tanks connected in series.		6
B)	What is Servo and Regulatory problem. Explain with Examples.		6
Q. 4	Solve Any Two of the following.		12
A)	Explain Routh-Hurwitz Criterion for stability.		6
B)	Explain Zeigler Nicholas method		6
C)	Determine the stability by Routh criterion for the system with Following characteristic equation $S^3+3S^2+5S+2=0$		6
Q.5	Solve Any Two of the following.		12
A)	Explain Bode Stability Criterion.		6
B)	Derive the transfer function of two interacting liquid level tanks connected in series.		6
C)	Consider a second order system with the following transfer function? $G(s)=Y(S)/X(S) = 1/ (S^2+S+1)$ Introduce a step change of magnitude one into the system and find (i)% Overshoot (ii) decay ratio (iii) Rise Time (iv) Ultimate Value of Y(t)		6
Q. 6	Solve Any Two of the following.		12
A)	Explain Bode Diagrams for first order systems.		6
B)	Explain Nyquist plots for first order, second order,pure dead-time and pure capacities system.		6
C)	Explain in details cascade control.		6
	*** End ***		